Advanced Functions & Equations

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Function composition**

Using the same f and g, find the formula for $(f \circ g)(x)$.

$(f \circ g)(x) = f(g(x)) = 3(x^2 + 1) - 2 = 3x^2 + 1.$

2 **[Foundational]** • **Target: Inverse functions**

Verify your answer above by computing $f(f^{-1}(5))$.

$f^{-1}(5) = \frac{5+2}{3} = \frac{7}{3}$, and $f\left(\frac{7}{3}\right) = 3\left(\frac{7}{3}\right) - 2 = 7 - 2 = 5$. It matches, confirming the inverse.

3 **[Foundational]** • **Target: Factoring higher-degree polynomials**

Given that $x = 1$ is a root of $x^3 - 2x^2 - 5x + 6 = 0$, use the Factor Theorem to factor completely and find all roots.

By the Factor Theorem, $(x - 1)$ is a factor. Dividing gives $x^3 - 2x^2 - 5x + 6 = (x - 1)(x^2 - x - 6) = (x - 1)(x - 3)(x + 2)$. Roots: $x = 1,\ 3,\ -2$.

4 **[Foundational]** • **Target: Factoring higher-degree polynomials**

Verify that $x = -2$ is a root of $x^3 - 2x^2 - 5x + 6 = 0$ by direct substitution.

$(-2)^3 - 2(-2)^2 - 5(-2) + 6 = -8 - 8 + 10 + 6 = 0$. Yes, $x = -2$ is a root.

5 **[Foundational]** • **Target: Exponential and logarithmic equations**

Solve $\log_3(x + 2) + \log_3(x - 2) = 2$, checking for extraneous roots.

Combine: $\log_3[(x + 2)(x - 2)] = 2 \Rightarrow x^2 - 4 = 3^2 = 9 \Rightarrow x^2 = 13 \Rightarrow x = \pm\sqrt{13}$. The domain requires $x - 2 > 0$, i.e. $x > 2$; since $\sqrt{13} \approx 3.61 > 2$ but $-\sqrt{13} < 2$, only $x = \sqrt{13}$ is valid.

6 **[Foundational]** • **Target: Function composition**

Find the slant (oblique) asymptote of the rational function: $g(x) = \frac{2x^3 - 5x^2 + 4x + 1}{x^2 - x + 1}$

Step 1: Perform polynomial long division: Dividing $2x^3 - 5x^2 + 4x + 1$ by $x^2 - x + 1$: 1. First term: $2x^3/x^2 = 2x$. Multiply: $2x(x^2 - x + 1) = 2x^3 - 2x^2 + 2x$. Subtract: $(-5x^2 + 4x) - (-2x^2 + 2x) = -3x^2 + 2x + 1$. 2. Second term: $-3x^2/x^2 = -3$. Multiply: $-3(x^2 - x + 1) = -3x^2 + 3x - 3$. Subtract: $(2x + 1) - (3x - 3) = -x + 4$ (remainder). Step 2: Equation of slant asymptote: The quotient is the linear equation: **$y = 2x - 3$**.

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7 [Foundational] • Target: Inverse functions

Find the vertical and horizontal asymptotes of the rational function: $f(x) = \frac{4x^2 - 1}{2x^2 + 5x - 3}$

Step 1: Factor numerator and denominator: $f(x) = \frac{(2x-1)(2x+1)}{(2x-1)(x+3)} = \frac{2x+1}{x+3} \quad (x \neq \frac{1}{2})$ Step 2: Vertical asymptote: The non-canceling factor in denominator is $x + 3 = 0 \Rightarrow \mathbf{x = -3}$. (The point at $x = 1/2$ is a removable hole, not a vertical asymptote). Step 3: Horizontal asymptote: Equal degrees (degree 2): ratio of leading coefficients is $y = \frac{4}{2} = \mathbf{2}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Rational equations

Evaluate the limit of the rational expression by factoring: $\lim_{x \to 3} \frac{x^3 - 27}{x^2 - 2x - 3}$

Step 1: Direct substitution check: $\frac{3^3 - 27}{3^2 - 2(3) - 3} = \frac{27 - 27}{9 - 6 - 3} = \frac{0}{0} \quad \text{(indeterminate)}.$ Step 2: Factor numerator using difference of cubes: $x^3 - 27 = (x - 3)(x^2 + 3x + 9)$ Step 3: Factor denominator: $x^2 - 2x - 3 = (x - 3)(x + 1)$ Step 4: Cancel the common factor $(x - 3)$ and evaluate.

9 [Intermediate] • Target: Factoring higher-degree polynomials

Find the inverse function $f^{-1}(x)$ for $f(x) = \frac{2x+3}{x-5}$ where $x \neq 5$.

Step 1: Swap x and y: $x = \frac{2y+3}{y-5}$ Step 2: Clear fraction and solve for y: $x(y - 5) = 2y + 3 \Rightarrow xy - 5x = 2y + 3\ xy - 2y = 5x + 3 \Rightarrow y(x - 2) = 5x + 3\ y = \mathbf{\frac{5x+3}{x-2}} \quad (x \neq 2).$

10 [Intermediate] • Target: Exponential and logarithmic equations

In triangle ABC, $a = 10$, $b = 16$, and $A = 30^\circ$. Determine whether $0, 1,$ or 2 distinct triangles can be formed, and calculate all possible values for angle B to the nearest tenth of a degree.

Step 1: Set up the Law of Sines for angle B: $\frac{\sin B}{b} = \frac{\sin A}{a} \Rightarrow \sin B = \frac{b \sin A}{a} = \frac{16 \sin 30^\circ}{10} = \frac{16(0.5)}{10} = \frac{8}{10} = 0.8.$ Step 2: Find the acute angle solution B_1: $B_1 = \sin^{-1}(0.8) \approx \mathbf{53.1}^\circ$. Step 3: Test the obtuse supplementary angle B_2: $B_2 = 180^\circ - 53.1^\circ = 126.9^\circ$. Check triangle sum with given $A = 30^\circ$: \[A + B_2 = 30^\circ + 126.9^\circ = 156.9^\circ Conclusion: Exactly **2** d.

11 [Intermediate] • Target: Function composition

Determine the value of the constant k for which the function $f(x)$ is continuous at $x = 2$: $f(x) = \{kx^2 + 2x, , x \leq 2; 3x + k, , x > 2$

Step 1: Apply the definition of continuity at a point (Definition 6.4): Continuity requires $\lim_{x \to 2^-} f(x) = \lim_{x \to 2^+} f(x) = f(2)$. Step 2: Evaluate the one-sided limits: Left-hand limit and function value: $\lim_{x \to 2^-} f(x) = f(2) = k(2)^2 + 2(2) = 4k + 4.$ Right-hand limit: $\lim_{x \to 2^+} f(x) = 3(2) + k = 6 + k.$ Step 3: Set them equal and solve for k: $4k + 4 = 6 + k \Rightarrow 3k = 2 \Rightarrow \mathbf{k = \frac{2}{3}}.$ Check: For $k = 2/3$.

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12 [Intermediate] • Target: Inverse functions

Solve the rational equation: $\frac{2}{x-3}+\frac{1}{x}=\frac{x-1}{x-3}$

Step 1: Domain restrictions: $xeq0$ and $xeq3$. Step 2: Multiply by $x(x-3)$: $2x+1(x-3)=x(x-1)$ $3x-3=x^2-x\Rightarrow x^2-4x+3=0$. Step 3: Factor: $(x-3)(x-1)=0\Rightarrow x=3$ or $x=1$. Step 4: Check domain: $x=3$ makes denominators zero (extraneous, rejected). Therefore, the only valid solution is $\mathbf{x=1}$.

13 [Intermediate] • Target: Rational equations

Find the value of constant k such that the function $f(x)$ is continuous at $x=3$:
$f(x)=\left\{\frac{x^2-9}{x-3},,xeq3;k,,x=3\right.$

Step 1: Continuity condition: $\lim_{x\to3}f(x)=f(3)=k$. Step 2: Evaluate the limit as $x\to3$:
$\lim_{x\to3}\frac{x^2-9}{x-3}=\lim_{x\to3}\frac{(x-3)(x+3)}{x-3}=\lim_{x\to3}(x+3)=3+3=6$. Conclusion: $\mathbf{k=6}$.

14 [Intermediate] • Target: Factoring higher-degree polynomials

The graph of $y=f(x)$ is transformed to $y=-2f(x-3)+5$. Describe the sequence of geometric transformations applied to the parent function.

Transformations in order: 1. Horizontal shift: Shift right by 3 units (due to $x-3$). 2. Vertical stretch: Stretch vertically by a factor of 2. 3. Reflection: Reflect across the x-axis (due to the negative sign). 4. Vertical shift: Shift upward by 5 units.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Exponential and logarithmic equations • Cross-Lesson: 3.01 Sequences & Series

If $f(x)=2x^2-3$ and $g(x)=\sqrt{x+1}$, find the domain of the composite function $(g\circ f)(x)$.

Step 1: Form composition: $(g\circ f)(x)=g(f(x))=\sqrt{(2x^2-3)+1}=\sqrt{2x^2-2}$. Step 2: Find domain by requiring radicand ≥0: $2x^2-2\geq0\Rightarrow x^2\geq1$. Step 3: Solve inequality: $\mathbf{(-\infty,-1]\cup[1,\infty)}$.

16 [Challenging] • Target: Function composition • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

Solve the exponential equation for x: $4^{x+1}=8^{2x-1}$

Step 1: Express both sides using base 2: $4=2^2\Rightarrow4^{x+1}=(2^2)^{x+1}=2^{2x+2}$
$8=2^3\Rightarrow8^{2x-1}=(2^3)^{2x-1}=2^{6x-3}$ Step 2: Equate exponents: $2x+2=6x-3\Rightarrow5=4x\Rightarrow\mathbf{x=\frac{5}{4}}$.

17 [Challenging] • Target: Inverse functions • Cross-Lesson: 6.01 Limits

[Test format & strategy · Practice 2] If the governing relation for this system is $g(t)=4t+4$, evaluate $g(6)$ and find the inverse input t when $g(t)=32$.

Substitute $t=6$: $g(6)=4(6)+4=\mathbf{28}$. Setting $4t+4=32$ gives $4t=28\Rightarrow t=\mathbf{7}$.

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18 [Challenging] • Target: Rational equations • Cross-Lesson: 1.01 Complex Numbers

Find all vertical and horizontal asymptotes, the exact coordinates of any holes, and all intercepts of the rational function $R(x) = \frac{2x^2+5x-3}{x^2+4x+3}$.

Factor numerator and denominator completely: $R(x) = \frac{(2x-1)(x+3)}{(x+1)(x+3)}$. The factor $(x+3)$ cancels for $xeq-3$. Evaluate the simplified form $g(x) = \frac{2x-1}{x+1}$ at $x=-3$: $g(-3) = \frac{2(-3)-1}{-3+1} = \frac{-7}{-2} = 3.5$. Key features:Hole: $(-3, 3.5)$Vertical asymptote: $x+1=0 \Rightarrow x=-1$Horizontal asymptote: Equal degrees (2 and 2) $\Rightarrow y = 2/1 = 2$$x$-intercept: Numerator root $2x-1=0 \Rightarrow (0.5, 0)$$y$-intercept.

19 [Challenging] • Target: Factoring higher-degree polynomials • Cross-Lesson: 2.01 Matrices & Vectors

Given $f(x) = \frac{2x+5}{x-3}$ for $xeq3$, find the inverse function $f^{-1}(x)$ and state its domain.

1. Set $y = \frac{2x+5}{x-3}$ and swap x and y: $x = \frac{2y+5}{y-3}$. 2. Multiply by $y-3$: $x(y-3) = 2y+5 \Rightarrow xy-3x = 2y+5$. 3. Collect y terms: $xy-2y = 3x+5 \Rightarrow y(x-2) = 3x+5$. 4. Divide by $x-2$: $f^{-1}(x) = \frac{3x+5}{x-2}$. The domain of $f^{-1}(x)$ excludes $x=2$: $(-\infty, 2) \cup (2, \infty)$. (This matches the range of f, which has horizontal asymptote $y=2$).

20 [Challenging] • Target: Exponential and logarithmic equations • Cross-Lesson: 3.01 Sequences & Series

Find the distance in the polar coordinate plane between the two points $P_1(4, \frac{\pi}{3})$ and $P_2(7, \frac{2\pi}{3})$.

Step 1: Apply the Law of Cosines for polar distance: $d = \sqrt{r_1^2 + r_2^2 - 2r_1r_2\cos(\theta_2 - \theta_1)}$ Step 2: Substitute values: $r_1 = 4$, $r_2 = 7$, $\theta_2 - \theta_1 = \frac{2\pi}{3} - \frac{\pi}{3} = \frac{\pi}{3}$. $d = \sqrt{4^2 + 7^2 - 2(4)(7)\cos\left(\frac{\pi}{3}\right)} = \sqrt{16 + 49 - 56\left(\frac{1}{2}\right)}$ $d = \sqrt{65-28} = \sqrt{\mathbf{37}}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Function composition

Given that $x=4$ is a root of $x^3 - 13x - 12 = 0$, use the Factor Theorem to find the other two roots.

a. $(-3, -1)$ ✓

b. $(3, 1)$

c. $(-2, -1)$

d. $(-3, -2)$

Dividing out $(x-4)$ gives $x^2 + 4x + 3 = 0$, which factors as $(x-(-1))(x-(-3)) = 0$, so $x=-1$ or $x=-3$. Answer: $(-3, -1)$.

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22 [Foundational] • Target: Inverse functions

A line L is tangent to the graph of the polynomial function $f(x) = x^4 - 6x^2 + 5x + 7$ at two distinct points in the xy-plane. What is the y-intercept of line L?

a. **-2 ✓**

b. 2

c. 7

d. 16

Let the equation of the tangent line L be $y = mx + b$. Because L is tangent to the quartic polynomial $f(x)$ at two distinct points, say $x = r$ and $x = s$ with res, each point of tangency corresponds to a root of multiplicity at least 2 for the difference equation $f(x) - (mx + b) = 0$. Since $f(x) - (mx + b)$ is a monic polynomial of degree 4, it factors completely over the real numbers as: $f(x) - (mx + b) = (x - r)^2(x - s)^2$ Expanding this product.

23 [Foundational] • Target: Rational equations

The polynomial function $P(x) = x^3 + ax^2 + bx + 12$, where a and b are real constants, has $x - 1$ as a factor. When $P(x)$ is divided by $x - 2$, the remainder is -10. What is the greatest root of the equation $P(x) = 0$?

a. **4 ✓**

b. 3

c. 1

d. 6

To find the roots of $P(x) = 0$, we first determine the unknown parameters a and b using the Factor Theorem and the Remainder Theorem. 1. **Apply the Factor Theorem:** Since $x - 1$ is a factor of $P(x)$, we have $P(1) = 0$: $P(1) = (1)^3 + a(1)^2 + b(1) + 12 = 0 \Rightarrow 1 + a + b + 12 = 0 \Rightarrow a + b = -13$ 2. **Apply the Remainder Theorem:** When $P(x)$ is divided by $x - 2$, the remainder is -10, so $P(2) = -10$.

24 [Foundational] • Target: Factoring higher-degree polynomials

The polynomial function $P(x) = 2x^3 + ax^2 - 5x + b$ has $(x - 1)$ as a factor. When $P(x)$ is divided by $(x + 2)$, the remainder is -18. What is the remainder when $P(x)$ is divided by $(x - 3)$?

a. **2 ✓**

b. 8

c. 18

d. -30

According to the Factor Theorem and the Remainder Theorem, if a polynomial $P(x)$ is divided by $(x - c)$, the remainder is $P(c)$. If $(x - c)$ is a factor, then $P(c) = 0$. Step 1: Set up the first equation using the factor $(x - 1)$: $P(1) = 2(1)^3 + a(1)^2 - 5(1) + b = 0$ $2 + a - 5 + b = 0 \Rightarrow a + b = 3$ Step 2: Set up the second equation using the divisor $(x + 2) = (x - (-2))$ with remainder -18: $P(-2) = 2(-2)^3 + a(-2)^2 - 5(-2) + b = -18$ $2(-8) + 4a + 10 + b = -18$.

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25 [Foundational] • Target: Exponential and logarithmic equations

In the standard (x, y) coordinate plane, the graph of $y = f(x)$ contains the point $(-2, 5)$. The function g is defined by $g(x) = f(x - 3) - 4$. Which of the following points must lie on the graph of $y = g(x)$?

a. $(1, 1)$ ✓

b. $(-5, 1)$

c. $(1, 9)$

d. $(-5, 9)$

We are given that the point $(-2, 5)$ lies on the graph of $y = f(x)$, which means $f(-2) = 5$. The function g is defined as $g(x) = f(x - 3) - 4$. To evaluate $g(x)$ using our known value $f(-2)$, we set the argument of f equal to -2: $x - 3 = -2 \Rightarrow x = 1$. Now, substitute $x = 1$ into the expression for $g(x)$: $g(1) = f(1 - 3) - 4 = f(-2) - 4 = 5 - 4 = 1$. Therefore, the point $(1, 1)$ must lie on the graph of $y = g(x)$. Geometrically, the transformation $y = f(x - 3) - 4$ translates the graph of.

26 [Foundational] • Target: Function composition

Given the functions $f(x) = 3x - 4$ and $g(x) = x^2 + 5$, what is the value of $f(g(2))$?

a. **23** ✓

b. 9

c. 18

d. 27

To find the value of the composite function $f(g(2))$, evaluate the inner function first, and then substitute that result into the outer function: 1. Evaluate $g(2)$: $g(2) = (2)^2 + 5 = 4 + 5 = 9$ 2. Evaluate $f(9)$: $f(9) = 3(9) - 4 = 27 - 4 = 23$ Common Misconceptions: - Evaluating in the reverse order, $g(f(2))$, yields $f(2) = 3(2) - 4 = 2$, and $g(2) = 2^2 + 5 = 9$. - Interpreting the expression as the product $f(2) \cdot g(2)$ yields $2 \cdot 9 = 18$. - Evaluating $3 \cdot g(2)$ but omitting the constant.

27 [Foundational] • Target: Inverse functions

Solve $\log_2 x + \log_2 (x - 3) = 2$, and give the valid solution.

a. **x=4** ✓

b. x=-1

c. x=4,\ x=-1

d. x=7

Combine: $\log_2 (x(x - 3)) = 2 \Rightarrow x(x - 3) = 2^2 = 4$. Factoring gives $x = 4$ or $x = -1$; reject $x = -1$ since $\log_2 x$ needs $x > 0$ (and $x - 3 > 0$ too). Valid answer: $x = 4$.

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28 [Foundational] • Target: Rational equations

Let $P(x)$ be a polynomial of minimum degree with real coefficients such that $\lim_{x\to 1} \frac{P(x)}{x^2-3x+2} = 5$ and $\lim_{x\to 2} \frac{P(x)}{x^2-3x+2} = -3$. What is the value of $P(3)$?

a. **-22** ✓

b. -11

c. 22

d. 42

First, factor the denominator: $x^2-3x+2=(x-1)(x-2)$. For $\lim_{x\to 1} \frac{P(x)}{(x-1)(x-2)}$ to exist and be finite when the denominator approaches 0, the numerator must also approach 0. Thus, $P(1)=0$, meaning $(x-1)$ is a factor of $P(x)$. Similarly, for $\lim_{x\to 2} \frac{P(x)}{(x-1)(x-2)}$ to exist, we must have $P(2)=0$, so $(x-2)$ is also a factor of $P(x)$. Therefore, $P(x)$ must be divisible by $(x-1)(x-2)$. We can write: $P(x)=(x-1)(x-2)Q(x)$ for.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Factoring higher-degree polynomials

The quadratic function $f(x)=ax^2+bx+c$ is graphed in the standard (x,y) coordinate plane below. The vertex of the parabola is at $(3,-6)$, and the graph passes through the point $(1,2)$. What is the value of b ?

a. -12 ✓

b. -6

c. 6

d. 12

1. Write the quadratic function in vertex form: $f(x)=a(x-h)^2+k$. 2. Given the vertex $(h,k)=(3,-6)$, the function is $f(x)=a(x-3)^2-6$. 3. Use the given point $(1,2)$ on the graph to find a: $2=a(1-3)^2-6\ 2=4a-6\Rightarrow 4a=8\Rightarrow a=2$. 4. Expand the equation to standard form ax^2+bx+c: $f(x)=2(x^2-6x+9)-6=2x^2-12x+18-6=2x^2-12x+12$. 5. Comparing coefficients gives $b=-12$. Common error: A student who ignores the scale factor a might cal.

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30 [Intermediate] • Target: Exponential and logarithmic equations

The function f is defined by $f(x) = \frac{ax+8}{cx+d}$ for all real numbers $x \neq -\frac{d}{c}$, where a, c, and d are nonzero constants. In the standard (x, y) coordinate plane, the graph of $y = f(x)$ has a horizontal asymptote at $y = 2$, a vertical asymptote at $x = 3$, and an x-intercept at $(-4, 0)$. What is the value of $f^{-1}(4)$?

a. **10** ✓

b. 16

c. -2

d. 2

We determine the constants a, c, and d step-by-step: 1. **Use the x-intercept:** The point $(-4, 0)$ lies on the graph, meaning $f(-4) = 0$. For a rational function, the numerator must equal zero at an x-intercept: $a(-4) + 8 = 0 \Rightarrow -4a = -8 \Rightarrow a = 2$ 2. **Use the horizontal asymptote:** For $f(x) = \frac{ax+8}{cx+d}$, as $x \to \pm\infty$, the horizontal asymptote is given by the ratio of the leading coefficients: $y = \frac{a}{c} = 2$ Since $a = 2$, we have.

31 [Intermediate] • Target: Function composition

A function f is defined for all real numbers x such that $x \neq 0$ and $x \neq 1$. If $f(x) + 2f\left(\frac{1}{1-x}\right) = 3x$ for all permissible values of x, what is the value of $f(3)$?

a. $\frac{20}{9}$ ✓

b. $\frac{7}{3}$

c. $\frac{16}{9}$

d. 3

To find $f(3)$, we use the cyclic substitution property of the transformation $T(x) = \frac{1}{1-x}$. 1. Substitute $x = 3$ into the given functional equation: $f(3) + 2f\left(\frac{1}{1-3}\right) = 3(3) \Rightarrow f(3) + 2f\left(-\frac{1}{2}\right) = 9 \quad - - - \text{ (Equation 1)}$ 2. Next, evaluate the equation at $x = -\frac{1}{2}$: $\frac{1}{1-(-1/2)} = \frac{1}{3/2} = \frac{2}{3}$.

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32 [Intermediate] • Target: Inverse functions

Consider the function $f(x) = \frac{x^3 - 8}{x^2 - 4}$. We are interested in the behavior of this function near its discontinuities. Let $x = a$ be a value where the function has a removable discontinuity, and let $x = b$ be a value where the function has a non-removable discontinuity (vertical asymptote). What is the value of the limit of $f(x)$ as x approaches a, and what is the value of $\lim_{x \to b^-} f(x)$?

a. Limit at a is 3, $\lim_{x \to b^-} f(x) = -\infty$ ✓

b. Limit at a is 3, $\lim_{x \to b^-} f(x) = \infty$

c. Limit at a is 1, $\lim_{x \to b^-} f(x) = -\infty$

d. Limit at a is 1, $\lim_{x \to b^-} f(x) = \infty$

First, we factor the numerator and the denominator of $f(x) = \frac{x^3 - 8}{x^2 - 4}$. The numerator is a difference of cubes: $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$. The denominator is a difference of squares: $x^2 - 4 = (x - 2)(x + 2)$. Thus, $f(x) = \frac{(x - 2)(x^2 + 2x + 4)}{(x - 2)(x + 2)}$. A removable discontinuity occurs where a factor cancels out. Here, $(x - 2)$ cancels, so the removable discontinuity is at $x = a = 2$. To find the limit as x approaches a, we consider the simplified function for $xeq2$.

33 [Intermediate] • Target: Rational equations

Consider the rational function $f(x) = \frac{ax^2 - 4x + 1}{x^2 - bx + 2}$. The graph of $y = f(x)$ has a vertical asymptote at $x = 1$ and a horizontal asymptote at $y = 2$. If $f(0) = \frac{1}{2}$, find the value of a.

a. 1

b. 2 ✓

c. 3

d. 4

For a rational function $f(x) = \frac{P(x)}{Q(x)}$, vertical asymptotes occur at the zeros of the denominator $Q(x)$ that are not also zeros of the numerator $P(x)$. Given a vertical asymptote at $x = 1$, we know that $x = 1$ is a root of the denominator $x^2 - bx + 2 = 0$. Therefore, $1^2 - b(1) + 2 = 0$, which simplifies to $1 - b + 2 = 0$, so $b = 3$. The denominator is $x^2 - 3x + 2 = (x - 1)(x - 2)$. The horizontal asymptote is determined by the ratio of the leading coefficients of the numerator and denominator if t.

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34 [Intermediate] • Target: Factoring higher-degree polynomials

Consider the rational function $f(x) = \frac{x^2 - ax + 6}{x - b}$. If the graph of $f(x)$ has a removable discontinuity at $x = 2$ and a vertical asymptote at $x = 3$, determine the values of a and b.

a. a = 5, b = 3 ✓

b. a = -5, b = 3

c. a = 5, b = 2

d. a = -5, b = 2

For a rational function to have a removable discontinuity at $x = c$, the numerator must have a factor of $(x - c)$ that cancels with a factor in the denominator. Given a removable discontinuity at $x = 2$, the numerator $x^2 - ax + 6$ must have $x = 2$ as a root. Thus, $2^2 - a(2) + 6 = 0$, which simplifies to $4 - 2a + 6 = 0$, so $10 - 2a = 0$, and $a = 5$. The numerator can be factored as $(x - 2)(x - 3)$ since $2 \times 3 = 6$ and $2 + 3 = 5$, so $a = 5$. For the function to have a vertical asymptote at $x = 3$, the.

35 [Intermediate] • Target: Exponential and logarithmic equations

The function f is defined for all real numbers $xeq -2$ by $f(x) = \frac{4x-1}{x+2}$ What is the value of $f^{-1}(3)$?

a. -5

b. 3

c. 5

d. 7 ✓

By definition of the inverse function, if $f^{-1}(3) = x$, then $f(x) = 3$. Set the function equal to 3 and solve for x: $\frac{4x-1}{x+2} = 3$ Multiply both sides by $x + 2$: $4x - 1 = 3(x + 2)$ $4x - 1 = 3x + 6$ Subtract $3x$ from both sides and add 1: $4x - 3x = 6 + 1$ $x = 7$ Thus, $f^{-1}(3) = 7$. Common Misconceptions: - Choice A (-5) arises from a sign error when clearing the denominator as $3(x - 2)$, yielding $4x - 1 = 3x - 6 \Rightarrow x = -5$. - Choice B (3) results from forgetting to dist.

36 [Intermediate] • Target: Function composition

Consider the rational function $f(x) = \frac{ax^2 + bx + c}{dx^2 + ex + f}$, where a, b, c, d, e, f are non-zero constants. Given that $\lim_{x \to \infty} f(x) = \frac{1}{2}$ and $\lim_{x \to -1} f(x) = \infty$, determine the value of $\frac{a}{d}$.

a. $\frac{1}{2}$ ✓

b. 1

c. 2

d. Cannot be determined

For a rational function where the degree of the numerator is equal to the degree of the denominator, the limit as $x \to \infty$ is the ratio of the leading coefficients. Therefore, $\lim_{x \to \infty} f(x) = \frac{a}{d}$. We are given that this limit is $\frac{1}{2}$, so $\frac{a}{d} = \frac{1}{2}$. The condition $\lim_{x \to -1} f(x) = \infty$ implies that $x = -1$ is a vertical asymptote. This means that the denominator is zero when $x = -1$, and the numerator is non-zero when $x = -1$. So.

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37 [Intermediate] • Target: Inverse functions

The graph of the function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of the parameter a.

a. 1

b. **2** ✓

c. 3

d. 4

We are given the function $f(x) = a \cdot b^x$ and two points it passes through: $(1, 12)$ and $(3, 108)$. Using the first point $(1, 12)$, we have: $12 = a \cdot b^1$ (Equation 1) Using the second point $(3, 108)$, we have: $108 = a \cdot b^3$ (Equation 2) To find a and b, we can divide Equation 2 by Equation 1: $\frac{108}{12} = \frac{a \cdot b^3}{a \cdot b^1}$ $9 = b^2$ Taking the square root of both sides, and assuming $b > 0$ for exponential functions: $b = 3$ Now, substitute the value of $b = 3$ back into E.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Rational equations • Cross-Lesson: 1.01 Complex Numbers

The function f is defined by $f(x) = -x^2 + bx + c$, where b and c are constants. In the standard (x, y) coordinate plane, the maximum value of $f(x)$ is 14, and $f(0) = 5$. If $b > 0$, what is the value of $f(5)$?

a. 5

b. 10 ✓

c. 13

d. 14

We are given $f(x) = -x^2 + bx + c$. Step 1: Determine the constant c. Evaluating at $x = 0$: $f(0) = -(0)^2 + b(0) + c = 5 \Rightarrow c = 5$ Step 2: Express the maximum value in terms of b. Since the coefficient of x^2 is -10, we have $b = 6$. Step 4: Evaluate $f(5)$. The complete function is $f(x) = -x^2 + 6x + 5$. $f(5) = -(5)^2 + 6(5) + 5 = -25 + 30 + 5 = 10$ A common error is confusing the parameter b with the vertex coordinate x_v (setting $x_v = 6$), which incorrectly leads to.

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39 [Challenging] • Target: Factoring higher-degree polynomials • Cross-Lesson: 2.01 Matrices & Vectors

Consider the rational function $f(x) = \frac{x^2 - 9}{x - 3}$. What is the value of $f(x)$ as x approaches 3?

a. 0

b. **6 ✓**

c. Undefined

d. 3

The function is given by $f(x) = \frac{x^2 - 9}{x - 3}$. We can factor the numerator as a difference of squares: $x^2 - 9 = (x - 3)(x + 3)$. Thus, for $x \neq 3$, we can simplify the function to $f(x) = \frac{(x - 3)(x + 3)}{x - 3} = x + 3$. To find the limit as x approaches 3, we can substitute 3 into the simplified expression: $\lim_{x \to 3} f(x) = \lim_{x \to 3} (x + 3) = 3 + 3 = 6$. Although the function is undefined at $x = 3$, the limit exists and is 6. Therefore, as x approaches 3, $f(x)$ approaches 6.

40 [Challenging] • Target: Exponential and logarithmic equations • Cross-Lesson: 3.01 Sequences & Series

Consider the exponential growth model described by the function $P(t) = P_0 e^{kt}$, where $P(t)$ is the population at time t, P_0 is the initial population, and k is the growth constant. If the initial population P_0 is 500 and the population after 4 years is 2000, determine the value of the growth constant k to three decimal places.

a. **0.347 ✓**

b. 0.416

c. 0.301

d. 0.477

The population model is given by $P(t) = P_0 e^{kt}$. We are given that $P_0 = 500$. We are also given that $P(4) = 2000$. Substituting these values into the equation, we get: $2000 = 500 e^{k \cdot 4}$ Divide both sides by 500: $4 = e^{4k}$ Take the natural logarithm of both sides: $\ln(4) = \ln(e^{4k})$ $\ln(4) = 4k$ Solve for k: $k = \frac{\ln(4)}{4}$ Using a calculator, $\ln(4) \approx 1.38629$. $k \approx \frac{1.38629}{4}$ $k \approx 0.34657$ Rounding to three decimal places, $k \approx 0.347$.

41 [Challenging] • Target: Function composition • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

Consider the rational function $f(x) = \frac{x^2 - 4}{x - 2}$. What is the value of the function as x approaches 2?

a. **4 ✓**

b. 0

c. Undefined

d. 2

The given function is $f(x) = \frac{x^2 - 4}{x - 2}$. We can factor the numerator as a difference of squares: $x^2 - 4 = (x - 2)(x + 2)$. Therefore, $f(x) = \frac{(x - 2)(x + 2)}{x - 2}$. For $x \neq 2$, we can cancel the $(x - 2)$ term, so $f(x) = x + 2$. As x approaches 2, the value of $f(x)$ approaches $2 + 2 = 4$. Although the function is technically undefined at $x = 2$ due to division by zero, the limit as x approaches 2 exists.

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42 [Challenging] • Target: Inverse functions • Cross-Lesson: 6.01 Limits

Consider the exponential function $f(x) = 2^x$. If $f(a) = 8$, what is the value of a?

a. 1

b. 2

c. **3** ✓

d. 4

The given function is $f(x) = 2^x$. We are given that $f(a) = 8$. Substituting a into the function, we get $2^a = 8$. To solve for a, we need to express 8 as a power of 2. Since $2^3 = 2 \times 2 \times 2 = 8$, we can rewrite the equation as $2^a = 2^3$. Since the bases are equal, the exponents must be equal. Therefore, $a = 3$.

43 [Challenging] • Target: Rational equations • Cross-Lesson: 1.01 Complex Numbers

Consider the exponential function $f(x) = 2^x$. What is the value of $f(3)$?

a. 6

b. **8** ✓

c. 9

d. 5

The function is given by $f(x) = 2^x$. To find $f(3)$, we substitute $x = 3$ into the function: $f(3) = 2^3$. Calculating the power, $2^3 = 2 \times 2 \times 2 = 8$. Therefore, the value of $f(3)$ is 8.

44 [Challenging] • Target: Factoring higher-degree polynomials • Cross-Lesson: 2.01 Matrices & Vectors

Consider the rational function $f(x) = \frac{x^2 - 9}{x - 3}$. What is the value of $f(4)$?

a. **7** ✓

b. 1

c. 5

d. 13

To find the value of $f(4)$, substitute $x = 4$ into the function. $f(4) = \frac{4^2 - 9}{4 - 3} = \frac{16 - 9}{1} = \frac{7}{1} = 7$.

45 [Challenging] • Target: Exponential and logarithmic equations • Cross-Lesson: 3.01 Sequences & Series

A 13 m ladder leans against a vertical wall and reaches 12 m up the wall. Find the angle the ladder makes with the ground, to the nearest tenth of a degree.

a. 42.7°

b. 22.6°

c. 72.4°

d. 67.4° ✓

In the right triangle, opposite $= 12$, hypotenuse $= 13$, so the angle is $\sin^{-1}\left(\frac{12}{13}\right) \approx 67.4^\circ$.

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